17.5 Gymnosperms

Gymnosperm are naked seed plant, i.e. the ovule is not enclosed by ovary. Gymnosperms have two phases in its life cycle (Gametophytic and Sporophytic). Sporophytic plant body is dominant and it is differentiated into root, stem and leaf. They have well developed vascular tissues (xylem and phloem). The water conducting tissue is tracheid and the food conducting tissue is sieve cell. They have cone in which sporangia and spores are produced.

17.5.1 Classification of Gymnosperms

Gymnosperm are classified into four different types. They are:

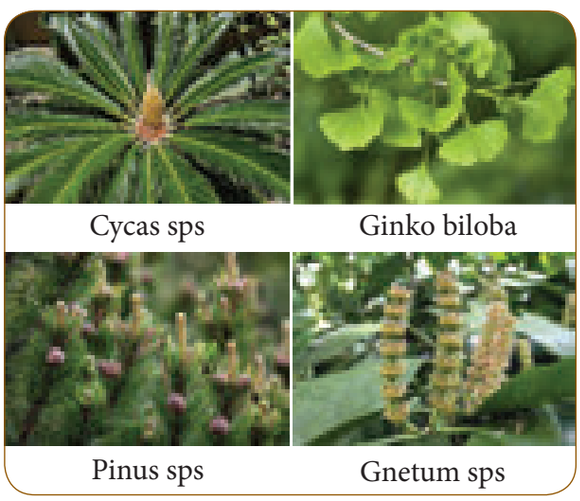
1. Cycadales
2. Ginkgoales
3. Coniferales
4. Gnetales

Cycadales

These are palm like small plants (erect and unbranched). Leaves are pinnately compound forming a crown. They have tap root system and coralloid root. E.g. *Cycas sps.*

Ginkgoales

These are large trees with fan shaped leaves. *Ginko biloba* is the only living species in the group. They produce unpleasant smell.



Cycas sps | Ginko biloba | Pinus sps | Gnetum sps

Figure 17.11 Gymnosperms

Coniferales

These are evergreen trees with cone like appearance. They have needle like leaves or scale leaves. Seeds are winged and produced in female cone. E.g. *Pinus sps.*

Gnetales

Gnetales are small group of plants. They possess advanced characters like angiosperm. Ovules are naked but, developed on flower like shoot. E.g. *Gnetum sps.*

17.5.2 Economic importance of Gymnosperms

- Woods of many conifers are used in the paper industries. E.g. *Pinus, Agathis*
- Conifers are the sources of soft wood for construction, packing and plywood industry. E.g. *Cedrus, Agathis*
- Turpentine, an essential oil, extracted from the resin of *Pinus* is used for paint preparation. It is also used medicinally to get relief from pain, bronchitis etc.,
- Seeds of *Pinus gerardiana* are edible.
- Ephedrine is an alkaloid extracted from *Ephedra*. It cures asthma and respiratory problems.
- *Araucaria bidwillii* is an ornamental plant.

17.6 Angiosperms

The term 'Angiosperm' is derived from two Greek words, i.e. 'angio' which means box or closed and 'sperma' which means seed. Habit of the plants may be herb *(Solanaum melongena),* shrub *(Hibiscus rosasinensis)* and tree (*Mangifera indica - Mango).* They have well developed vascular tissues called xylem and phloem. Xylem contains vessel, tracheid, xylem parenchyma and xylem fibre. Phloem contains sieve tubes, phloem parenchyma, companion cells and phloem fibres.

17.6.1 Classification of Angiosperms

Angiosperms are divided into two classes. They are:

- Dicotyledons
- Monocotyledons

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